Issa Cissé

Engineering student at École polytechnique (3rd year) — Applied Mathematics
Looking for a machine learning / artificial intelligence internship, starting 22 March 2027
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EDUCATION

École polytechnique | Engineering degree, 3rd year | Palaiseau, France | Sep. 2024 – present

- Main courses: Probability theory for ML (Monte Carlo methods, generative models), Signal processing: from Fourier to Machine Learning, Operations research, Modelling and simulation of climate risks, Markov chains, Control and optimisation, Data analysis with Python.

Institut national polytechnique Félix Houphouët-Boigny (INP-HB) | Preparatory classes | Yamoussoukro, Côte d'Ivoire | Sep. 2022 – Jul. 2024

- Intensive scientific preparation (mathematics, physics, chemistry) for the French grandes écoles entrance exams.

EXPERIENCE

Meditect – Yako | AI developer, part-time | Remote | Sep. 2026 – present

- Designed and led the technical side of a digital health platform, on the web and on **WhatsApp (Turn.io)**: **LLM-assisted teleconsultation** with handover to a doctor, mental health support (PHQ-9 / GAD-7 scores), women's health.
- Specified the **medical guardrails** around the model: emergency detection by fixed rules before the AI, critical computations on the server, prescriptions validated by a doctor.
- **Accounts and security**: phone verification by WhatsApp code, PIN, family profiles; **security audit** and fix of 4 vulnerabilities (access control, dictionary attack, code abuse); ~240 automated tests.
- **Put into production** three smartphone health scans (anaemia, pulse from video) on **Google Cloud Run**; traced a result mismatch of a **1D CNN** between two servers.
- AI-assisted development: design, review, testing and deployment of the code.

Meditect | Applied research intern (R&D) | Abidjan, Côte d'Ivoire | Jun. – Aug. 2026

- Estimated **anaemia risk from a photo of the eye** (conjunctiva), for dark-skinned populations underrepresented in the literature.
- Designed and ran a **field data collection**: 152 patients, 2 pharmacies, 6 weeks; trained a nurse on the protocol.
- Ran the **experiments and critical analysis of results**: colour feature selection, model comparison, **AUC ≈ 0.73** in nested cross-validation; detected and fixed labelling errors in the data.
- Full Python pipeline (segmentation, classification, web service) developed with AI assistance.

ACADEMIC PROJECT

Connected weather station | Group science project | Sep. 2025 – Apr. 2026

- Automatic protection of an observatory dome: rain and temperature sensors on a Raspberry Pi, real-time monitoring.
- Web interface for data visualisation and remote control.

AWARD

- 2nd prize, National Mathematics Competition (final year of high school) | Côte d'Ivoire

SKILLS

Machine learning: classification, cross-validation, metrics (AUC), data collection and quality, LLMs (prompting, tools, structured outputs, guardrails)
Application security: authentication, access control, abuse limits, attack tests
Programming: Python, Java; AI-assisted development
Tools: Google Cloud Run, Oracle Cloud, Turn.io (WhatsApp)
Languages: French (native), English (B1), Spanish (A2)

OTHER

Outreach: science mediation, Association Science Ouverte (Drancy): maths and astronomy workshops, guidance for high-school students | 2024 – 2025
Interests: Basketball, reading, music